Extruder Machine



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Cold Extrusion Machine; Hot Extruder Machine

What it is used for

An extruder machine forces a prepared material through a shaped die to produce a continuous form, such as food products, pellets, or other process-specific outputs. Correct material preparation, temperature and feed control, die selection, and safe shutdown help maintain product consistency and reduce pressure, heat, and entanglement hazards.

Main parts

- Feed hopper receives raw material.
- Feed screw or conveying mechanism moves and compresses material.
- Barrel contains the material during processing.
- Heating or cooling zones control material temperature where fitted.
- Drive motor and gearbox power the screw or extrusion mechanism.
- Die shapes the material as it exits the machine.
- Temperature sensors monitor barrel and die conditions.
- Control panel regulates speed, temperature, and operating settings.
- Cutting or discharge mechanism handles the extruded product where fitted.
- Guards and emergency stop help protect the operator.

Before you start

1. Only trained personnel should operate the extruder machine.
2. Read the manual and confirm whether the machine is configured for cold extrusion, hot extrusion, or food processing.
3. Verify that the raw material, moisture level, feed rate, die, and operating parameters are suitable for the product.
4. Inspect the hopper, barrel, die, drive, guards, controls, sensors, and electrical connections for damage.
5. Confirm that the die and downstream equipment are correctly installed and securely fastened.
6. For food applications, clean and sanitise product-contact parts and verify that ingredients and processing conditions meet the approved food safety procedure.
7. Wear suitable PPE, including heat-resistant gloves for hot processes, eye protection, and other protection required by the risk assessment.

How to use

1. Position the machine on a stable base and ensure the discharge area is clear and ready.

2. Install the correct die and any approved cutting or downstream attachments with the machine isolated.
3. Prepare and measure the raw material according to the approved process recipe.
4. Check that guards are fitted, emergency stop functions, and the hopper and discharge path are unobstructed.
5. Set the feed rate, screw speed, die configuration, and temperature zones according to the validated procedure.
6. For hot extrusion, allow the barrel and die to reach the specified operating temperature before feeding material.
7. Start the drive according to the manufacturer's sequence and verify that the screw operates normally.
8. Feed material gradually and monitor motor load, pressure indicators, temperature, and product flow.
9. Adjust only permitted parameters using the approved procedure and remain alert for blockages, unusual noise, or overheating.
10. Collect or cut the extruded product using the designed discharge system and keep hands clear of moving parts.
11. If pressure rises unexpectedly, material stops flowing, or a fault occurs, stop feeding and follow the manufacturer's safe shutdown and pressure-release procedure.
12. At completion, stop feeding and clear the machine only as specified by the manufacturer.
13. Switch off, isolate power, and allow hot components and material to cool or depressurise before disassembly and cleaning.
14. Clean, inspect, and record production results, material changes, and any faults.

⚠ Safety

- Never reach into the hopper, barrel, die, or cutter while the machine is operating.
- Do not loosen a die, flange, or barrel connection while material is hot or the system remains pressurised.
- Keep guards, interlocks, and emergency stops in working condition; never bypass them.
- Avoid contact with hot barrels, dies, and freshly extruded material; use suitable handling tools.
- Isolate electrical and mechanical energy before clearing blockages or servicing the screw and drive system.
- For food extrusion, prevent cross-contamination and follow the approved allergen, sanitation, and temperature-control procedures.

Care & storage

- Clean the hopper, die, barrel-accessible surfaces, and product-contact components using the approved procedure after safe shutdown.
- Inspect the screw, die openings, seals, bearings, drive coupling, and guards for wear or damage.
- Check temperature sensors, controllers, and pressure indicators at the recommended intervals.
- Lubricate designated drive components according to the manufacturer's instructions and keep lubricant away from food-contact areas.

- Store dies and removable parts clean and dry, and arrange authorised service for persistent pressure, temperature, or product-quality problems.

Common mistakes to avoid

- Using an incorrect die or unsuitable material preparation.
- Feeding too quickly or exceeding the machine's rated capacity.
- Opening a hot or pressurised assembly to clear a blockage.
- Cleaning or disassembling the machine before isolating power and allowing pressure and temperature to reach safe levels.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.